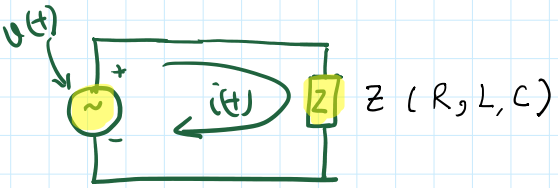
Lecture 16: Single Phase AC Circuits (Contd...)

RLC Circuits with AC Excitation

$v(t) = \sqrt{2}\, V \sin \omega t$

$i(t) = \sqrt{2}\, I \sin(\omega t - \phi)$



For Resistive Load:

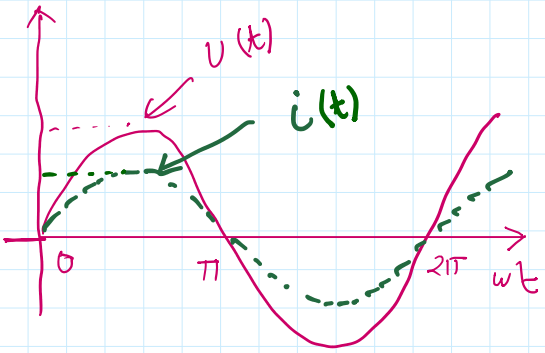
$$i = \frac{v}{R} = \frac{\sqrt{2}V}{R} \sin \omega t$$

$$= \sqrt{2}\, I \sin \omega t \; ; \; I = \frac{V}{R}$$

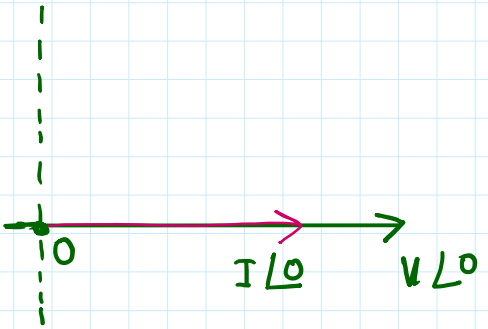
Using phasors, $\bar{V} = V \angle 0$

$$\bar{I} = \frac{\bar{V}}{R} = I \angle 0$$

Thus $I = \frac{V}{R}$, $\phi = 0^\circ$



voltage and current waveforms



Phasor diagram

Power:

$$\bar{S}_R = \bar{V}\, \bar{I}^* = V \angle 0 \; I \angle 0$$

$$= VI = P_R + jQ_R$$

$\Rightarrow \bar{S}_R = P_R$; $Q_R = 0$ (No reactive power in R)

$pf = \cos(0^\circ) = 1.0$

Instantaneous Power, $p(t)$:

$$p(t) = v(t)\, i(t) = \frac{v^2}{R}$$

$$= \frac{(\sqrt{2}V)^2 \sin^2 \omega t}{R} = \frac{V^2}{R}(2 \sin^2 \omega t)$$

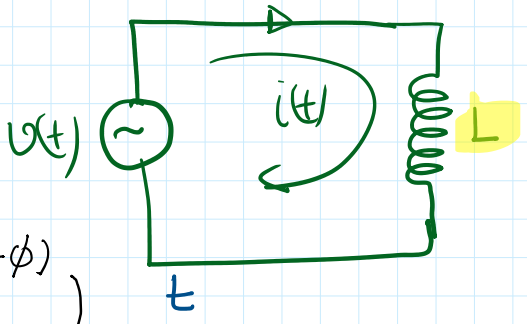
$p(t) = \frac{V^2}{R}(1-\cos 2\omega t)$ (which was earlier defined as active power)

Previous derivation: $p(t) = VI\cos\phi(1-\cos 2\omega t) - VI\sin\phi\sin 2\omega t$

Inductive Load :

$v(t) = V_m \sin \omega t = \sqrt{2}\, V \sin \omega t$

$i(t) = I_m \sin(\omega t + \phi) = \sqrt{2}\, I \sin(\omega t + \phi)$



Now $v_L = v(t) = L\frac{di}{dt}$

$$\text{Im}\left(V_m e^{j\omega t}\right) = L\frac{d}{dt}\,\text{Im}\left(I_m e^{j(\omega t+\phi)}\right)$$

$$= \text{Im}\left[L\frac{d}{dt}\left(I_m e^{j\phi} e^{j\omega t}\right)\right]$$

$$= \text{Im}\left[L\frac{d}{dt}\left(\bar{I}_m e^{j\omega t}\right)\right]$$

$$= \text{Im}\left(L\bar{I}_m e^{j\omega t}\cdot j\omega\right)$$

$$= \text{Im}\left(j\omega L\bar{I}_m e^{j\omega t}\right)$$

$$= \frac{d}{dt}\left[L\cdot\text{Im}\left(\bar{I}_m e^{j\omega t}\right)\right]$$

$$= \text{Im}\left[L\frac{d}{dt}\left(\bar{I}_m e^{j\omega t}\right)\right]$$

$$= \text{Im}\left(L\cdot\bar{I}_m (j\omega t) e^{j\omega t}\right)$$

$$\text{Im}\left(\bar{V}_m e^{j\omega t}\right) = \text{Im}\left[j\omega L\,\bar{I}_m e^{j\omega t}\right]$$

$$\text{Im}\left(\frac{\bar{V}_m}{\sqrt{2}} e^{j\omega t}\right) = \text{Im}\left(j\omega L\frac{\bar{I}_m}{\sqrt{2}} e^{j\omega t}\right)$$

$$\Rightarrow \bar{V} = (j\omega L)\,\bar{I} = \bar{I}\,Z_L$$

where Z_L = Impedance of inductance $L = j\omega L$

where Z_L = Impedance of inductance, $L = j\omega L$

$= j 2\pi f L$

X_L = Reactance of inductance $= \omega L = 2\pi f L$

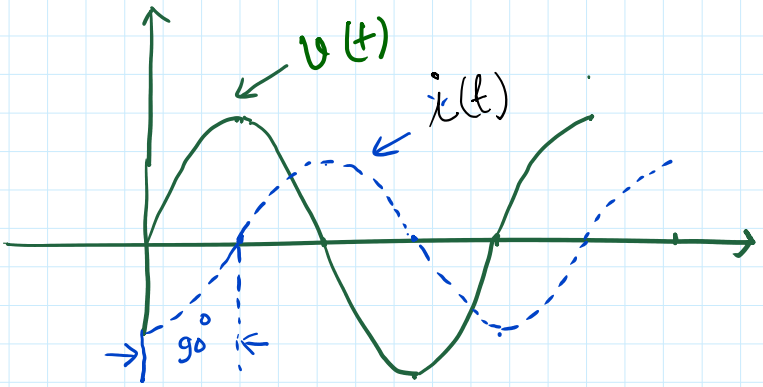
Thus $\bar{I} = \bar{I}_L = \frac{\bar{V}_L}{j\omega L} = \frac{\bar{V}}{Z_L} = \frac{V\angle 0}{\omega L \angle 90^\circ} = \frac{V}{\omega L}\angle -90^\circ$

$= I\angle -90^\circ$

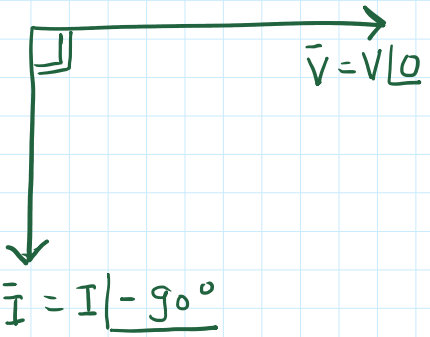
Therefore $I = \frac{V}{\omega L} = \frac{V}{|Z_L|}$

$\phi = -90^\circ$

$i(t) = \frac{\sqrt{2}V}{\omega L} \sin(\omega t - 90^\circ)$ for $v(t) = \sqrt{2}\, V \sin \omega t$



Voltage and current waveforms for inductor load



Phasor diagram:

Powers of Inductor:

$p_L(t) = v_L(t)\, i_L(t)$

$= V_m \sin \omega t \cdot I_m \sin(\omega t - 90^\circ)$

$= \frac{V_m I_m}{2}\{2 \sin \omega t \cdot \sin(\omega t - 90^\circ)\}$

$= V_m I_m [\cos 90^\circ - \cos(2\omega t - 90^\circ)]$

$$= \frac{V_m I_m}{2}\left[\cos 90^\circ - \cos(2\omega t - 90^\circ)\right]$$

$$p_L(t) = -VI\sin 2\omega t$$

Compare with the general expression:

$$p(t) = VI\cos\phi(1-\cos 2\omega t) - VI\sin\phi\sin 2\omega t$$

Now $\phi = 90^\circ \Rightarrow p_L(t) = -VI\sin 2\omega t$

(Reactive Power due to reactive element, L)

Powers Using Phasors:

$$\bar{S} = \bar{V}_L \bar{I}_L^{\,*} = V\angle 0 \cdot (I\angle -90^\circ)^* = VI\angle 90^\circ$$

$$= VI\cos 90^\circ + jVI\sin 90^\circ$$

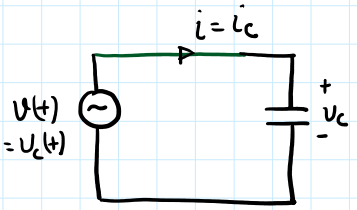
$$= 0 + jVI$$

$$\Rightarrow P_L = 0\ W$$

$$Q_L = VI = V\cdot\frac{V}{X_L} = \frac{V^2}{X_L} = I^2X_L\ VAr$$

$$pf = \cos(-90^\circ) = 0\ (\text{lagging})$$

A Capacitive Circuit:



$v(t) = \sqrt{2}V\sin\omega t = V_m\sin\omega t$

$i(t) = \sqrt{2}I\sin(\omega t+\phi)$

$= I_m\sin(\omega t+\phi)$

$$v_C = \frac{1}{C}\int i\,dt$$

$$\downarrow$$

$$\mathrm{Im}\left(V_m e^{j\omega t}\right) = \frac{1}{C}\int \mathrm{Im}\left[I_m e^{j(\omega t+\phi)}\right]dt$$

$$\text{Im}\left(V_m e^{j\omega t}\right) = \frac{1}{C}\int \text{Im}\left[I_m e^{\cdots}\right] dt$$

$$= \text{Im}\int \frac{I_m e^{j\phi}\cdot e^{j\omega t}}{C}\, dt$$

$$\text{Im}\left(\bar{V}_m e^{j\omega t}\right) = \text{Im}\left[\frac{\bar{I}_m}{j\omega C} e^{j\omega t}\right]$$

$$\Rightarrow \bar{V}_m = \frac{\bar{I}_m}{j\omega C} \quad \text{or} \quad \bar{V} = \frac{\bar{I}}{(j\omega C)} = \bar{I}\cdot\left(\frac{1}{j\omega C}\right) = \bar{I} Z_c$$

$$Z_c = \frac{1}{j\omega C} = -\frac{j}{\omega C} = -jX_c \; ; \; X_c = \text{reactance of the capacitance}$$

$$\boxed{X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}}$$

$$\bar{I} = \frac{\bar{V}}{Z_c} = \frac{V\angle 0}{\left(\frac{1}{j\omega C}\right)} = V\angle 0 * j\omega C$$

$$= V.\omega C \angle 90^\circ \qquad j = 1\angle 90^\circ = e^{j\pi/2}$$

$$\boxed{\bar{I} = \frac{\bar{V}}{|Z|}\angle 90^\circ}$$

$$i(t) = \frac{\sqrt{2}V}{|Z|}\sin(\omega t + 90^\circ)$$

$$= C\frac{dV_c}{dt} = C\, V_m(\cos\omega t)\,\omega$$

$$= \frac{V_m}{(1/\omega C)}\sin(\omega t + 90^\circ)$$

$$\bar{S} = \bar{V}\bar{I}^* = P + jQ$$

$$= V\angle 0 * \frac{V}{(1/\omega C)}\cdot\angle -90^\circ$$

$$= V^2\omega C\cdot\angle -90^\circ = V^2\omega C * 0 - jV^2\omega C$$

$$\bar{S} = -jV^2\omega C \Rightarrow P = 0,\; Q = -V^2\omega C$$

$$\boxed{Q = -\frac{V^2}{X_c}}$$

$pf = 0$ (leading)

$P = VI\cos\phi$

$Q = VI\sin\phi$

$= -VI$

$= -V^2\omega C$

$= -V^2/X_c$

